

Motion Encoding for Human Perceptual Similarity

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Abstract

We ask whether a learned representation of human motion can predict *perceptual* similarity—how similar two movements appear to human observers—more faithfully than geometric distance measures (quaternion/6D rotation geodesic, Dynamic Time Warping) that dominate motion analysis. Using Laban-effort-annotated motion-capture performances of three action types (walking, pointing, picking) paired with a triplet-based human similarity-rating dataset, we show that a self-supervised **masked motion predictor augmented with an auxiliary pose-reconstruction objective (MAMP+pose)** produces motion embeddings whose **raw, unsupervised** distances align with human perception more strongly than both geometric baselines *and* a generic learned (reconstruction-only SSL) motion encoder across all three actions, under a strict, repeated, leakage-clean nested cross-validation—to our knowledge the first single encoder to do so, and notably including the expressively sparse pointing action where time-warping baselines were expected to dominate. A progressive ablation isolates the contribution of each objective and motivates the final design: reconstruction objectives capture static configuration but never model dynamics; **masked motion prediction** alone is the complement—recovering the picking action it best serves but collapsing on walking and pointing; and **composing** masked motion prediction with an auxiliary **pose-reconstruction** objective supplies both capabilities in one encoder, the only configuration competitive across all three actions. We further show that the human ratings, when used to **fine-tune the encoder** under a perceptual objective (rather than to fit a metric on a fixed embedding), further improve held-out alignment on all three actions. All claims are reported as fold-averaged correlations with calibrated variability, against baselines re-evaluated on the identical folds.

Keywords

motion similarity, self-supervised learning, perceptual similarity, Laban Movement Analysis, masked motion prediction

1 Introduction

1.1 Motion similarity and geometric distances as measures of similarity

Quantifying how *similar* two human movements are is foundational to motion retrieval, synthesis, style transfer, and the perceptual study of movement. The dominant approaches are geometric distances computed on raw kinematic features:

- **Rotation geodesic distance**—the distance on the rotation group $SO(3)$ between corresponding joint orientations, aggregated over joints and time. It respects the manifold structure of rotations but assumes (after temporal alignment) frame-wise correspondence.

- **Dynamic Time Warping (DTW)**—a non-linear temporal alignment that factors out speed variation and emphasizes the *shape* and *progression* of a gesture. DTW is the most widely adopted geometric framework for time-series motion comparison precisely because it models the temporal structure to which human observers are sensitive.

Though strong, well-motivated baselines, the scientific question is, whether a *learned* motion representation can predict human perceptual judgments better than they do.

1.2 Contributions

- (1) A **single self-supervised motion encoder (MAMP+pose)** whose **raw, unsupervised** embedding distances beat both geometric baselines (DTW, rotation geodesic) **and a generic learned (reconstruction-only SSL) motion encoder** on human-perceptual similarity across all three action types—including the expressively sparse pointing action—when evaluated on motion clips held out of pretraining, under a repeated, leakage-clean nested cross-validation with all baselines re-evaluated on the identical folds. The learned baseline shows that *merely* learning a motion representation is insufficient (it does not uniformly beat geometry); the composition is what wins all three. To our knowledge this is the first single encoder to do so on this heterogeneous action set, and it requires **no perceptual supervision and no metric-learning stage**.
- (2) A **mechanistic, progressive-ablation account** of *why* the masked-motion-plus-pose composition is the right encoder: reconstruction objectives capture static configuration but not dynamics; masked motion prediction is the complement (a picking specialist that collapses on walking and pointing); and only composing the two yields an encoder competitive across walking, pointing, and picking. Each step's result motivates the next design choice, evidence-first.
- (3) A demonstration that the human ratings are productively used by **fine-tuning the encoder** under a perceptual objective derived from the comparison study—an **anchored ranking loss whose margin is the human dissimilarity gap itself** (hyperparameter-free and metric-aligned), which further improves held-out alignment on all three actions.
- (4) A **methodological contribution**: a repeated, leakage-clean nested cross-validation protocol—with baselines re-evaluated on the identical folds—needed to make reliable, reproducible perceptual-similarity claims on a small, high-variance human-rating set, and a characterization of the single-split pitfalls it corrects.

2 Data corpus

2.1 Stimulus space: Laban effort tuples

Stimuli are motion-capture performances parameterized by Laban Movement Analysis (LMA) **effort tuples** $(e_1, e_2, e_3, e_4) \in \{-1, 0, +1\}^4$, encoding the four Laban effort dimensions (Space, Time, Weight, Flow). Following established LMA semantics, the number of non-zero efforts has perceptual meaning: 0 non-zero = *neutral* (1 class); 2 non-zero = *state* (24 classes); 3 non-zero = *drive* (32 classes); single-effort and full-effort tuples are rare in natural motion and are excluded.

The perceptual study uses the **neutral + state + drive** subset: **57 effort classes per action** (1 neutral + 24 states + 32 drives), performed for three action types—**walking, pointing, picking**—for a total of 171 effort-class stimuli. Each performance additionally has a left/right-mirrored twin; the LMA states/drives clips excluded from encoder pretraining (Section 2.3) comprise $57 \times 3 \times 2 = 342$ clips.

2.2 Motion representation

Each clip is converted to a per-frame feature of **34 joints $\times$ 6-D rotation representation**: joint index 0 carries the root position/orientation, indices 1–33 carry 6-D continuous joint rotations. The 6-D representation (Gram–Schmidt-recoverable to $SO(3)$) is preferred over quaternions for its continuity and gradient stability.

Each of the 204 feature channels (34×6) is **z-score standardized** using a per-channel mean and standard deviation computed once over the entire training corpus; values are then clipped to ± 10 . The same fixed statistics are applied at training and evaluation, so no evaluation-clip information enters the normalization.

Fixed-length windowing. The encoder operates on a fixed **120-frame** window (four seconds at 30 fps). Each clip is mapped to this length deterministically: clips longer than 120 frames are truncated to their first 120 frames; shorter clips are **right-padded by repeating their final frame**. This identical rule is applied in pretraining, in embedding extraction, and in held-out encoding—there is no length-dependent branching between train and test. (The geometric baselines of Section 5.2, by contrast, operate on the *raw, unpadded* sequences.)

Clip-length heterogeneity. Walking and picking clips are long (median 73 and 71 frames; up to 137), whereas pointing clips are markedly shorter (median 53 frames; max 100). Because short clips are tail-padded, pointing clips contribute the largest fraction of repeated, frozen-pose frames under the 120-frame window—a consequence of their brevity that connects to their limited kinematic expressiveness (Section 3).

2.3 Pretraining corpus (self-supervised encoder)

The encoder is pretrained self-supervised on a broad, heterogeneous motion corpus of **5,795 BVH clips** drawn from three sources: the LMA effort performances (433 clips, including the eval actions); the Bandai-Namco Research Motion Dataset [Kobayashi et al., 2023] (3,077 clips of professionally captured everyday actions in multiple expressive styles); and the CMU Graphics Lab Motion Capture Database (2,285 general-purpose performances).

A single common skeleton. The three sources were originally captured with different marker sets, joint hierarchies, frame rates, and file conventions. A substantial preprocessing effort re-targeted **all** sources onto one canonical skeleton—the **CMU joint hierarchy**—and re-expressed every clip as a uniform BVH file at a common **30 fps**. This re-targeting is what makes the corpus trainable by a single encoder: every clip yields the identical **34-joint $\times$ 6-D rotation** per-frame feature (Section 2.2), regardless of source.

Held-out construction. To guarantee a **strict held-out evaluation**, all 342 LMA neutral/state/drive clips (mirror twins included)—exactly the clips on which perceptual similarity is evaluated—are excluded from pretraining (and from the encoder’s own validation), leaving $\sim 5,453$ training clips. The encoder therefore never observes any evaluation stimulus during representation learning. For the corpus-fair comparison against TMR (Section 6.3), we additionally train a **corpus-matched** MAMP+pose whose pretraining corpus is size-matched to TMR’s HumanML3D training budget.

2.4 Human perceptual ratings (ground truth)

Similarity ratings were collected via **triplet (2-of-3) comparison trials**. In each trial a participant viewed three motions—**Left (0), Neutral (1), Right (2)**—and selected the two judged most similar. The dataset comprises **1,540 triplet trials per action** (one per distinct non-neutral effort pair, with the neutral as the third item), each rated by a **median of 11 participants** (range 10–18); presentation order was randomized. The three pairings $\{(0, 1), (0, 2), (1, 2)\}$ of a given triplet received normalized choice frequencies summing to 1, interpretable as probabilities of selecting each pairing as most-similar.

2.4.1 Notation for the three choice frequencies. Fix a triplet with **Left (0)** and **Right (2)** the two effort exemplars and **Neutral (1)** the neutral exemplar. Write the three aggregated choice frequencies (each in $[0, 1]$, summing to 1) as $c(0, 2)$ (Left–Right chosen), $c(0, 1)$ (Left–Neutral), and $c(1, 2)$ (Neutral–Right). $c(0, 2)$ is the quantity of primary interest: how often observers judged the two effort exemplars *mutually most similar*, an absolute measure of the perceived similarity of that effort pair on a common $[0, 1]$ scale.

2.4.2 The evaluation target: a perceptual dissimilarity in $[0, 1]$. For each non-neutral effort pair we define the perceptual **dissimilarity** as

$$d_{\text{perc}} = 1 - c(0, 2) \in [0, 1].$$

A pair almost always grouped together ($c(0, 2) \rightarrow 1$) has $d_{\text{perc}} \rightarrow 0$; a pair almost never grouped together has $d_{\text{perc}} \rightarrow 1$. **d_{perc} is the single ground-truth quantity against which every distance measure—the learned embedding distances and both geometric baselines—is correlated** (Section 5.2). It uses only the absolute Left–Right frequency and no triplet-internal contrasts, so it is method-agnostic.

2.4.3 Perceptual training signals. The perceptual fine-tuning of Section 6.4 requires a per-pair supervisory signal derived from the triplets. The **reported method trains on the ordinal d_{perc} relation** via a ranking loss; a finer, neutral-referenced construction—the **dynamic alpha**—is one of the tested alternatives, defined here. Per non-neutral effort pair we generate two directed Left–Right alphas

contrasting the direct Left–Right frequency against each neutral-mediated alternative:

$$\alpha(0 \rightarrow 2) = c(0, 2) - c(0, 1), \quad \alpha(2 \rightarrow 0) = c(0, 2) - c(1, 2).$$

Each alpha lies in $[-1, +1]$; a positive alpha is evidence the pair should sit *close* in embedding space, a negative alpha that it should sit *far*. These alphas are used only by the neutral-anchored *alternative* arms, which Section 6.4 finds do not generalize. A note on hygiene: the reported ranking method supervises on the *ordinal relation* induced by d_{perc} , while evaluation scores the Spearman correlation between embedding distances and d_{perc} on the held-out test fold; because train, selection, and test folds are disjoint (Section 5.2), no class’s rating is ever both a training target and its own test.

2.5 Metric of merit

The ground truth is **ordinal** by construction (a ranking induced by choice frequencies) and the learned metric optimizes ordinal triplet inequalities; **Spearman rank correlation is the primary metric**. Pearson is reported as a magnitude-alignment supplement.

3 Per-action motion structure

The three actions differ markedly in *kinematic expressiveness*—how much of the body moves, over how large a range, and over how long. We quantify this from the motion data, averaging per-effort-class statistics over the 57 classes of each action (28 articulated joints; temporal “activity” = a joint whose across-frame standard deviation exceeds 0.01 in the standardized feature space).

Table 1: Per-action kinematic descriptors (mean over effort classes).

Descriptor	Walking	Pointing	Picking
Active joints (of 28)	24.6	11.6	20.0
Active-joint fraction	0.88	0.41	0.71
Median clip length (frames)	73	53	71
Mean per-joint temporal std	0.089	0.048	0.090
Mean range of motion	0.276	0.123	0.290

Pointing is, by every measure, the least kinematically expressive action. On average only ~12 of 28 joints (41%) move meaningfully in a pointing clip, versus ~25 (88%) for walking and ~20 (71%) for picking; pointing’s per-joint variability and range of motion are roughly *half* those of the other two actions, and its clips are the shortest. This expressive sparsity leads us to two hypotheses for pointing’s distinctive difficulty: (**H1, signal density**) fewer varying joints give the encoder less structure from which to separate pointing effort classes, so raw pointing geometry should be less discriminative; and (**H2, temporal brevity**) shorter clips afford the masked-motion pretext task fewer dynamic regions to model. Under both, the prediction is the same: pointing’s *raw* encoder distances will be the least discriminative of the three actions. We see this borne out (Section 6): pointing is the lowest of MAMP+pose’s three raw correlations and the highest-variance action. What is not foreordained is whether that residual signal is nevertheless enough to beat the geometric baselines; we find that with the right pretraining objective it is.

4 Encoder design—a progressive ablation toward MAMP+pose

The encoder design was not chosen a priori; it is the endpoint of a sequence of controlled training runs, each motivated by the failure of the previous. We present that sequence because it establishes the central **two-capability composition** finding—that perceptual similarity across heterogeneous actions requires an encoder modeling *both* motion dynamics and static pose configuration—and rules out the obvious single-objective alternatives.

We evaluate every encoder identically: pool its output to one vector per clip, take L2 distances between clips, and correlate against d_{perc} under the repeated, leakage-clean nested cross-validation of Section 5.2 (all Section 4 numbers are fold-averaged Spearman, $n = 25$). **No encoder is evaluated at a hand-chosen training epoch.** Within each fold the training epoch is an inner-loop hyperparameter: a candidate checkpoint is taken on a **uniform 100-epoch grid across the model’s full trajectory**, the checkpoint maximizing raw perceptual Spearman on the disjoint inner (SELECT) fold is chosen, and only then is it scored on the untouched TEST fold. The grid is identical for every model, removing checkpoint-selection as a researcher degree of freedom. Encoders differ along two axes we deliberately separate—the **autoencoder backbone** (Section 4.1) and the **training objective** (Sections 4.2–4.3).

4.1 Step 1—backbone: the MLD

SkipTransformer over a plain transformer

We consider two transformer-autoencoder backbones for the reconstruction encoders: a **plain transformer autoencoder** (a standard encoder/decoder with a compressed latent bottleneck) and the **MLD-style transformer autoencoder** of Motion Latent Diffusion [Chen et al., CVPR 2023]—a transformer encoder/decoder with **U-Net-like long skip connections** (“SkipTransformer”), symmetric long-range links from each early encoder layer to the matching late decoder layer. The skips carry high-frequency per-frame detail forward that the plain bottleneck discards. Holding the *objective* fixed (reconstruction-only) and varying *only* the architecture:

Table 2: Reconstruction-only encoders, raw Spearman ($n = 25$, mean $\pm$ SE).

Encoder	Walking	Pointing	Picking
Plain transformer AE	+0.476 $\pm$ 0.031	+0.213 $\pm$ 0.034	+0.491 $\pm$ 0.031
MLD SkipTransformer AE	+0.532 $\pm$ 0.023	+0.331 $\pm$ 0.039	+0.498 $\pm$ 0.031

The MLD SkipTransformer wins on **all three actions at the same objective**, decisively on pointing (+0.118), consistent with the skips preserving high-frequency pose detail that the plain bottleneck discards.

Deterministic vs. variational. We use the **deterministic** MLD autoencoder rather than its variational form (MLD-VAE) on principled grounds, not performance: a VAE adds a KL term toward a prior (a generative design choice) that confounds the reconstruction probe, and our headline encoder is itself deterministic, so a deterministic comparator is apples-to-apples. Empirically the two

are comparable on the perceptually decisive pointing axis (+0.331 AE vs. +0.324 VAE); MLD-VAE achieves the strongest walking score in the sweep (+0.597) but remains a walking/picking specialist that stalls on pointing.

4.2 Step 2—objective: a controlled velocity-vs-reconstruction factorial

Having fixed the backbone, we ask whether an explicit **velocity** term endows the encoder with dynamics. To avoid arbitrary loss weighting, we run a **controlled single-variable factorial**: each loss term is on (unit weight) or off, on **both** backbones.

Table 3: Velocity factorial, raw Spearman ($n = 25$, mean $\pm$ SE).

Encoder (objective)	Walking	Pointing	Picking
<i>Plain transformer</i>			
reconstruction-only	+0.476 $\pm$ 0.031	+0.213 $\pm$ 0.034	+0.491 $\pm$ 0.029
reconstruction + vel	+0.472 $\pm$ 0.026	+0.205 $\pm$ 0.039	+0.399 $\pm$ 0.037
velocity-only	+0.321 $\pm$ 0.033	+0.362 $\pm$ 0.036	+0.471 $\pm$ 0.033
<i>MLD SkipTransformer</i>			
reconstruction-only	+0.532 $\pm$ 0.023	+0.331 $\pm$ 0.039	+0.498 $\pm$ 0.029
reconstruction + vel	+0.537 $\pm$ 0.022	+0.339 $\pm$ 0.035	+0.493 $\pm$ 0.037
velocity-only	+0.451 $\pm$ 0.031	+0.278 $\pm$ 0.031	+0.254 $\pm$ 0.034

Three results. **(1) Adding a velocity term to reconstruction changes almost nothing**—flat within noise on MLD, flat-to-worse on the plain transformer; on neither backbone does it move pointing across the +0.370 bar. **(2) Velocity-only is a weaker objective overall and reshapes which action is served**—with reconstruction removed, global quality drops (MLD picking collapses +0.498 $\rightarrow$ +0.254). **(3) The telling signal**: velocity-only is the only reconstruction-family cell that lifts plain-transformer pointing, and it lifts it markedly (+0.213 $\rightarrow$ +0.362), while degrading walking. We are careful about what this licenses—it does not tell us where pointing’s perceptual signal physically resides—but it does establish that a purely temporal target carries perceptual signal a pose target does not surface, a first indication that **dynamics information is perceptually relevant** and under-exploited by reconstruction. Yet velocity as an added *loss* cannot capitalize on it (result 1). The lesson is therefore not “add velocity” but “capture dynamics properly,” motivating the move to a task that *forces* dynamics into the representation.

4.3 Step 3—why a velocity loss cannot supply dynamics

A velocity loss on a visible reconstruction is **partly redundant**. A reconstruction autoencoder sees the entire clip and is rewarded for reproducing every frame; if it reconstructs the poses x_t accurately, the frame-to-frame differences $x_{t+1} - x_t$ —the velocities—are reproduced automatically. A velocity term is therefore largely already satisfied and supplies little additional gradient. This predicts precisely what the factorial shows. The lesson that carries forward: **dynamics must be forced into the representation by the learning task itself**—by making the model *predict* motion it cannot see—not appended as a loss term on a fully-visible reconstruction.

4.4 Step 4—masked motion prediction, not reconstruction

The resolution comes from the Masked Motion Prediction (MAMP) framework [Mao et al., ICCV 2023]. Rather than reconstructing visible poses, MAMP **masks ~80% of spatio-temporal patches and predicts the motion (temporal deltas) of the masked regions** from sparse visible context, with a motion-magnitude prior biasing masking toward dynamically rich regions. This imposes genuine dynamics-modeling pressure: the encoder cannot copy what it cannot see.

MAMP alone is a dynamics specialist. The motion-only MAMP encoder’s single strong action is picking (+0.490 raw, the one bar it clears), while it is weak on walking (+0.351) and pointing (+0.299). What is solid—and load-bearing—is the **empirical complementarity** between the two objectives: masked-motion prediction succeeds on exactly the action (picking) where reconstruction was weakest, and fails on exactly the actions (walking, pointing) where reconstruction was strongest. Neither objective alone serves all three, which motivates *composing* them.

4.4.1 Pretraining budget and the pretext-vs-downstream divergence. We adopt the original MAMP pretraining hyperparameters and pretrain to convergence of the self-supervised loss, saving the full checkpoint trajectory; the reported epoch is selected per fold on the inner SELECT fold over the uniform 100-epoch grid. For the MAMP family this consistently selects an **early** checkpoint—the held-out perceptual signal rises quickly and plateaus by a few hundred epochs while the self-supervised loss keeps descending—a concrete instance of the pretext-vs-downstream divergence. Selection is leakage-clean (SELECT fold picks; TEST fold reports).

4.5 Step 5—the pose augmentation (MAMP+pose)

Masked motion prediction supplies dynamics but discards the static-configuration fidelity that reconstruction recovered—and with it walking and pointing. We augment it with an **auxiliary pose-reconstruction objective**. In standard MAMP a single linear head maps each masked patch’s decoder feature to its **motion** target (the temporal delta $x_{t+s} - x_t$); we add a **second, parallel linear head** off the *same* decoder features mapping each masked patch to its **pose** target (the raw per-frame value x_t). The training loss is $L = L_{\text{motion}} + \lambda_{\text{pose}} \cdot L_{\text{pose}}$ with $\lambda_{\text{pose}} = 1.0$. Because both heads read the same decoder features and backpropagate into the same encoder, the encoder is pressured to produce one latent that simultaneously encodes how the body is *moving* (the dynamics that win picking) and how it is *configured* (the static fidelity walking and pointing depend on). Two parallel heads (rather than summing targets) let each target be predicted in its natural form—a temporal delta and an absolute pose live on different scales—while sharing all representation-bearing layers. Section 6.5 shows this composition rescues walking and pointing while retaining picking.

5 The perceptual inference pipeline

5.1 Raw-embedding regime (unsupervised)

A clip is mapped to the fixed 120-frame window, standardized, and patchified into spatio-temporal patches—one patch per (4-frame

block $\times$ joint), giving $30 \times 34 = 1020$ patches. The pretrained encoder (run with mask ratio 0, all patches visible) produces a 256-D feature per patch; we mean-pool to a single **256-D clip embedding** (the encoder-native linear-probe readout). Pairwise **Euclidean (L2) distances** between clip embeddings are correlated (Spearman) against d_{perc} over the held-out test subset. This regime uses *no human ratings* during representation learning and therefore measures the encoder's intrinsic perceptual alignment.

5.2 Evaluation protocol and baselines

Repeated, leakage-clean nested cross-validation. All reported correlations are the mean over **5 repeats $\times$ 5 outer folds** ($n = 25$). Effort classes are partitioned (magnitude-stratified) into 5 folds with a repeat-specific seed. For each outer fold: the fold is the **test** set (reported, never seen during training or selection); one disjoint fold is the **selection** set, used *only* to pick each encoder's checkpoint by raw held-out Spearman; the remaining folds are the **training** set. The encoder additionally never saw these clips during pretraining, so the test fold is held out of both stages. We report mean $\pm$ standard error ($\text{SE} = \text{SD}/\sqrt{25}$).

Why not a single fixed split. The per-action correlation is strongly split-dependent, most severely for pointing, whose value ranges roughly $[0.23, 0.55]$ depending only on which classes land in the held-out subset. Conclusions drawn from one split do not survive re-partitioning; the repeated nested-CV averages over this variance and is the basis for every claim.

Baselines on identical folds. Both geometric baselines are computed on the **raw, unpadded** motion features over the **same 25 test folds** and correlated against the same d_{perc} . The per-frame geodesic cost maps each joint's 6-D feature to $R \in \text{SO}(3)$ and takes the geodesic angle $\theta = \arccos((\text{tr}(R_1^\top R_2) - 1)/2)$, averaged over the 33 articulated joints; DTW finds the monotone alignment minimizing accumulated per-frame geodesic cost (length-normalized). The "geodesic" baseline is the DTW-aligned mean per-frame cost on the same raw unpadded sequences. (*Correction to a legacy implementation:* an earlier version padded clips to fixed length by repeating the final frame and compared frame-wise without alignment; for short pointing clips this injected long runs of frozen-pose frames, spuriously deflating distances. Computing on raw unpadded sequences with proper alignment removes this bias and *strengthens* the baseline.)

6 Results

6.1 Headline—held-out, repeated nested cross-validation

We report Spearman correlation between each method's pair distances and human dissimilarity d_{perc} . All headline numbers are the mean over the repeated, leakage-clean nested cross-validation; the geometric baselines are computed on the identical 25 folds and fold-averaged the same way.

On Spearman, the **raw, unsupervised MAMP+pose encoder beats both geometric baselines and the learned baseline on all three actions** under matched held-out evaluation—to our knowledge the first single encoder to do so. Treating each fold-difference as paired across the 25 matched folds, the pointing and picking advantages over the geometric baselines are large and well-separated

Table 4: Raw MAMP+pose vs. geometric and learned baselines ($n = 25$, mean $\pm$ SE).

Method	Walking	Pointing	Picking
MAMP+pose, raw	+0.552 $\pm$ 0.026	+0.434 $\pm$ 0.030	+0.543 $\pm$ 0.033
DTW (geometric)	+0.490 $\pm$ 0.028	+0.273 $\pm$ 0.039	+0.303 $\pm$ 0.038
Geodesic (geometric)	+0.371 $\pm$ 0.028	+0.234 $\pm$ 0.031	+0.264 $\pm$ 0.030
Vanilla SSL (recon-only)	+0.476 $\pm$ 0.031	+0.213 $\pm$ 0.034	+0.491 $\pm$ 0.029

from zero ($\approx 3-5\sigma$ on a conservative independent-SEM test); the one near-marginal cell is walking vs. DTW ($+0.062$, $\approx 1.6\sigma$), though walking clears geodesic by a wide margin. The all-three result is a property of the raw encoder geometry and does not require any learned metric refinement.

6.2 The raw-encoder result—a stronger, unsupervised claim

The all-three win is achieved by the **raw** encoder embeddings—no perceptual supervision whatsoever. The perceptual-similarity structure is therefore *intrinsic to the self-supervised representation*, not imposed by a downstream metric. Beating a heavily-optimized temporal baseline like DTW with unsupervised embeddings—including on the expressively sparse pointing action, where DTW's time-warping was expected to dominate—indicates the masked-motion pretext task, augmented with the pose head, genuinely captures the kinematic structure underlying human similarity judgments.

6.3 Comparison to a supervised learned representation (TMR)

A sterner test is a **purpose-built, supervised** motion representation. We compare against **TMR** [Petrovich et al., ICCV 2023], a text $\leftrightarrow$ motion retrieval model trained with contrastive language supervision on HumanML3D—a representation shaped by a *large* corpus of semantic annotations, the opposite supervision regime to ours.

Protocol and a conservative handicap. TMR consumes HumanML3D 263-D motion features, so we convert each rated clip from BVH to SMPL parameters (joints2smpl SMPLify fit, mean reconstruction error ≈ 4 cm $\approx 2\%$ of body height) and then to the 263-D features TMR expects, encode through the released HumanML3D TMR motion encoder, and evaluate on the **identical 25 nested-CV folds** and the same d_{perc} . The ≈ 4 cm conversion perturbs TMR's input slightly, so any MAMP+pose advantage is a **conservative lower bound**. For a corpus-fair comparison we use the corpus-matched MAMP+pose (Section 2.3) as the comparator.

Table 5: Corpus-matched MAMP+pose vs. TMR ($n = 25$, mean $\pm$ SE).

Method	Walking	Pointing	Picking
TMR (supervised, HumanML3D)	+0.452 $\pm$ 0.025	+0.425 $\pm$ 0.036	+0.408 $\pm$ 0.031
MAMP+pose (matched, raw)	+0.462 $\pm$ 0.027	+0.302 $\pm$ 0.031	+0.481 $\pm$ 0.031
+ ranking fine-tune	+0.699 $\pm$ 0.022	+0.456 $\pm$ 0.036	+0.605 $\pm$ 0.031

Two honest reads. **Raw**, the corpus-matched MAMP+pose beats TMR on walking and picking but **TMR is stronger on pointing** (+0.425 vs. +0.302)—the one action where abundant semantic supervision helps most, and where our raw encoder is weakest. TMR is the strongest pointing baseline in the entire study (far above DTW's +0.273) and, unlike the geometric baselines, does not collapse on pointing—it is roughly flat across actions, consistent with a semantically supervised representation capturing broad motion structure uniformly but not tuned to perceptual effort-similarity. **With the light ranking fine-tune (Section 6.4), MAMP+pose beats TMR on all three actions**, including pointing (+0.456 vs. +0.425), achieved with a far smaller supervision set ($\approx 1,540$ within-task perceptual triplets per action) than TMR's corpus of semantic annotations. The finding is one of *alignment over abundance*: sparse supervision aligned to the target task matches or exceeds abundant supervision aligned to a different task. We state the pointing result precisely—TMR wins pointing on the raw encoder and is edged only after the perceptual fine-tune—rather than overclaiming a raw all-three win against this baseline.

6.4 Perceptual fine-tuning: using the human ratings to improve the encoder

The raw encoder already wins. We now ask whether the small budget of human ratings can push it further. The productive way, we find, is to back-propagate a perceptual objective **into the encoder itself**, reshaping the representation so that embedding distances track human dissimilarity—rather than fitting a separate metric on a frozen embedding.

An anchored ranking loss with a preference-scaled margin. The ground truth d_{perc} is *ordinal*, so the natural objective is a **ranking** loss. For an anchor class i and two others j, k , let $p_j = d_{\text{perc}}(i, j)$ and $p_k = d_{\text{perc}}(i, k)$; the human ordering names a *near* and a *far* class. With pairwise distances $\hat{d}$ min-max normalized to $[0, 1]$ within the batch, we impose

$$L = \text{relu}(\hat{d}(i, \text{near}) - \hat{d}(i, \text{far}) + m), \quad m = |p_k - p_j|,$$

i.e. the anchor–near distance must fall below the anchor–far distance by a margin equal to the **human dissimilarity gap** itself. Two properties make this the principled choice: the margin is **hyperparameter-free and adaptive** (confidently-ordered pairs are pushed apart harder than near-ties, with no margin constant to tune), and the loss is **metric-aligned by construction**—it optimizes the same ordinal quantity, over the same non-neutral pairs, that the Spearman evaluation scores.

Leakage-clean protocol and a single held-out selection. Fine-tuning runs under the same repeated nested-CV as the raw results; the only change is that within each fold's training portion the MAMP+pose encoder is fine-tuned rather than frozen. Two design points make the reported numbers conservative. First, the base checkpoint is chosen *per fold on the selection fold only*. Second, we use a **single** held-out selection: rather than also early-stopping on the selection fold, we train a **fixed budget and report the final epoch**—removing any “selection used twice” concern by construction.

The ranking fine-tune **lifts all three actions substantially** from the corpus-matched raw base. It is robust to the stopping rule:

Table 6: Ranking fine-tune vs. raw MAMP+pose ($n = 25$, mean $\pm$ SE).

Method	Walking	Pointing	Picking
raw MAMP+pose (no ratings)	+0.462 $\pm$ 0.027	+0.302 $\pm$ 0.031	+0.481 $\pm$ 0.028
+ ranking fine-tune	+0.699 $\pm$ 0.022	+0.456 $\pm$ 0.036	+0.605 $\pm$ 0.025

reporting the fixed-budget final epoch matches—in fact slightly exceeds—a variant that additionally early-stops on the selection fold, so the single-selection design loses nothing. (A direct-regression objective and a neutral-anchored, alpha-margin objective were also tried; both under-perform the ranking loss, the latter failing to generalize on pointing—consistent with the perceptual signal residing in the direct inter-state ordering the ranking loss targets rather than in neutral-referenced context shifts.) In short, the human ratings are best used not as a metric fit on a fixed representation but as a **light, ranking-based fine-tuning of the encoder**.

6.5 Ablation—the pose head is necessary

Table 7: Motion-only MAMP vs. MAMP+pose ($n = 25$, mean $\pm$ SE).

Objective	Walking	Pointing	Picking
motion-only (MAMP)	+0.351 $\pm$ 0.032	+0.299 $\pm$ 0.032	+0.490 $\pm$ 0.028
+ pose head (MAMP+pose)	+0.552 $\pm$ 0.026	+0.434 $\pm$ 0.030	+0.543 $\pm$ 0.025

Motion-only masked prediction clears the bar only on picking, its best action, and is weak on walking and pointing. Adding the pose head **lifts every action**, most sharply walking (+0.351 $\rightarrow$ +0.552) and pointing (+0.299 $\rightarrow$ +0.434, the gain that carries it clear of the baselines), while retaining picking. This confirms the two-capability composition: the motion target supplies dynamics, the pose target supplies configuration fidelity, and the combination is the only encoder clearing both baselines on all three actions.

6.6 Summary of the per-action story

A single self-supervised encoder, with no perceptual supervision, beats both geometric baselines on all three actions under repeated nested cross-validation—including the expressively sparse pointing action. Where the human ratings help further, it is by fine-tuning the encoder (Section 6.4), which lifts all three actions above the already-baseline-beating raw encoder.

6.7 Full model sweep (all encoders, raw)

All values are from the same repeated nested cross-validation ($n = 25$), evaluated on the canonical 57 effort classes per action; each encoder's checkpoint is selected per fold on the disjoint selection fold from an 11-checkpoint uniform 100-epoch grid. The geometric bars each encoder must clear (the stronger baseline per action) are **walking** > 0.478 , **pointing** > 0.370 , **picking** > 0.431 ; we mark $\checkmark$ when the encoder's 95% CI lower bound (mean -1.96 -SE) clears that bar.

Table 8: Raw embeddings, Spearman ($n = 25$, mean $\pm$ SE).

Encoder (backbone, objective)	Walking	Pointing	Picking	all 3?
Plain, recon-only	+0.476 $\pm$ 0.031	+0.213 $\pm$ 0.034	+0.491 $\pm$ 0.029	Encoder
Plain, recon+vel	+0.472 $\pm$ 0.026	+0.205 $\pm$ 0.039	+0.399 $\pm$ 0.037	MAMP (plain enc, motion-only)
Plain, velocity-only	+0.321 $\pm$ 0.033	+0.362 $\pm$ 0.036	+0.471 $\pm$ 0.033	MAMP-uencfull (U-Net enc, motion-only)
MLD AE, recon-only	+0.532 $\pm$ 0.023	+0.331 $\pm$ 0.039	+0.498 $\pm$ 0.029	MAMP+pose (plain enc, +pose)
MLD AE, recon+vel	+0.537 $\pm$ 0.022	+0.339 $\pm$ 0.035	+0.493 $\pm$ 0.030	—
MLD AE, velocity-only	+0.451 $\pm$ 0.031	+0.278 $\pm$ 0.031	+0.254 $\pm$ 0.039	—
MLD VAE, recon-only	+0.597 $\pm$ 0.021	+0.324 $\pm$ 0.032	+0.501 $\pm$ 0.023	—
MAMP (motion-only)	+0.351 $\pm$ 0.032	+0.299 $\pm$ 0.032	+0.490 $\pm$ 0.033	—
MAMP+pose	+0.552$\pm$0.026	+0.434$\pm$0.030	+0.543$\pm$0.032	—

MAMP+pose is the only encoder whose 95% CI clears all three baseline bars. Pointing is the discriminator: the strongest competitors (the MLD AEs) reach walking/picking comfortably but stall on pointing at +0.32–0.34 (CI not clearing 0.370), exactly the expressively-sparse action the composition is built to capture. No competing encoder is strong on all three actions; each alternative is a specialist. The architecture and objective axes are separable and both matter: at a fixed reconstruction objective the MLD backbone beats the plain transformer, while at a fixed backbone it is the masked-motion-plus-pose objective—not reconstruction with or without a velocity term—that produces the only all-three winner. Section 6.8 tightens this into a direct de-confound.

6.8 Disentangling architecture from objective—is it the pose head or the backbone?

The step from the reconstruction encoders to MAMP changes **two** things at once—the *objective* (reconstruction $\rightarrow$ masked prediction) and, implicitly, the *backbone*, since the MLD reconstruction encoders carry U-Net skip connections while the MAMP encoder is a plain (skip-free) transformer. A skeptic could argue that MAMP+pose recovers configuration not because of the pose *objective* but because skip connections—the mechanism that lifts reconstruction pointing from +0.213 to +0.331—would do the same for masked prediction. We test this directly.

An MLD-style encoder inside MAMP. We port the MLD Skip-Transformer’s *encoder-internal* U-Net into the MAMP encoder, holding the masked-motion objective, corpus, normalization, and split fixed. Because MAMP masks $\approx 80\%$ of tokens before the encoder, we run the U-Net encoder on the **full, unmasked** sequence and route only the visible-position features to the decoder, so the encoder is train/inference-consistent and its skips operate on complete motion while the prediction task remains non-trivial (masked targets never leak to the decoder). We call this **MAMP-uencfull**; crucially it has **no pose head**, isolating the effect of the MLD-style *architecture* under the masked objective.

The result is unambiguous on the discriminating axis. Giving MAMP the MLD-style U-Net encoder **does not recover pointing**: it moves from +0.299 to +0.275—statistically flat, if anything slightly lower—and remains below even the MLD reconstruction encoders’ pointing and far below the bar. The pose *objective*, by contrast, lifts pointing to +0.434 (≈ 4 SE gain). **The configuration recovery that**

Table 9: Architecture vs. objective on the masked-prediction backbone ($n = 25$, mean $\pm$ SE).

	Walking	Pointing	Picking
MAMP (plain enc, motion-only)	+0.351 $\pm$ 0.032	+0.299 $\pm$ 0.032	+0.490 $\pm$ 0.033
MAMP-uencfull (U-Net enc, motion-only)	+0.470 $\pm$ 0.029	+0.275 $\pm$ 0.028	+0.566 $\pm$ 0.023
MAMP+pose (plain enc, +pose)	+0.552$\pm$0.026	+0.434$\pm$0.030	+0.543$\pm$0.032

carries pointing is supplied by the pose-reconstruction objective, not by the encoder architecture. Two secondary observations sharpen the mechanism: the U-Net encoder is not inert—it lifts the coarse-motion actions (walking +0.351 $\rightarrow$ +0.470, and the best picking in the sweep, +0.566)—it simply does not manufacture the fine configuration signal the sparse pointing action requires; and we run it on the full, unmasked sequence deliberately, as the architecture’s best case, since skips taken after masking (on the visible $\approx 20\%$ only) are uniformly weaker.

We complete the 2×2 of architecture $\times$ objective, since MLD’s own recipe couples both skip connections and a reconstruction objective. The remaining cell—an MLD-style U-Net encoder trained *with* the pose objective—asks whether the two compound.

Table 10: Architecture $\times$ objective (raw Spearman, $n = 25$, mean $\pm$ SE).

	plain encoder	MLD-style U-Net encoder
motion-only	MAMP: +0.299 $\pm$ 0.032	uencfull: +0.275 $\pm$ 0.028
+ pose	MAMP+pose: +0.434 $\pm$ 0.030	[pending]

[Placeholder—the MAMP-uencfull+pose cell is pending a training run; to be filled with the $n = 25$ mean $\pm$ SE on completion. If its pointing exceeds +0.434, architecture and objective are complementary—the U-Net encoder’s benefit unlocks only with the pose objective, echoing MLD’s own coupling of skips with reconstruction; if $\approx +0.434$, the pose objective saturates configuration recovery and the architecture adds nothing on top. Either outcome does not alter the conclusion that the pose objective, not the backbone, recovers configuration.]

7 Why MAMP+pose succeeds: modeling vs. copying

The central mechanistic claim is the distinction between **modeling** dynamics and **copying** poses. Reconstruction objectives present the model with the entire visible motion and reward reproducing it; under full visibility, the velocities are reproduced for free as a byproduct of accurate pose copying, so a velocity term is partly redundant. Velocity-penalized reconstruction thus remains a **pose-biased** representation. Masked motion prediction removes the copying shortcut: with $\sim 80\%$ of patches hidden and the motion of the hidden, high-energy regions as the target, the encoder must infer kinematic progression it cannot observe—it must internalize how the body moves, not merely where it currently is. This is why motion-only masked prediction is the empirical complement of reconstruction. The auxiliary pose head restores the static-configuration fidelity reconstruction had. The **composition**—masked dynamics modeling plus pose reconstruction—supplies both capabilities in one

encoder and is the only configuration competitive across all three heterogeneous actions; neither objective alone is.

8 Reproducibility

8.1 Evaluation protocol (exact)

Repeated nested cross-validation. 5 repeats $\times$ 5 outer folds ($n = 25$). Magnitude-stratified folds per action with seed 42+1000·repeat. Per outer fold f : test = fold f ; selection = fold $(f+1) \bmod 5$; training = remaining three folds. The neutral exemplar is added to every subset. Checkpoint selection uses raw held-out Spearman on the selection fold (never the test fold). For the perceptual fine-tuning the selection fold is used *once*, to pick the per-fold base checkpoint; the fine-tune itself runs a **fixed epoch budget** (no second, early-stopping use of the selection fold).

Evaluation universe. The canonical 57 effort classes per action. The stored embedding directories also contain non-canonical clips (single-effort and fully-polarized tuples) carrying *no* human ratings; these cannot enter any correlation, but if left in the pool that is shuffled to form the split they perturb which rated classes land in each fold and silently change every number—most severely for pointing. The split must shuffle the canonical 57 only. (An earlier inability to reproduce the pointing baseline traced entirely to this.)

Per-pair target. $d_{\text{perc}} = 1 - \text{count_normalized}[(\text{selected}_0=0, \text{selected}_1=2)]$. Correlation is Spearman of pooled-embedding L2 distance vs. d_{perc} over exactly the test-fold pairs that carry a human rating, using plain pairwise L2 with no neutral-centering.

Strict hold-out and single corpus. The 342 LMA evaluation clips are excluded from pretraining and its validation; all reported encoders are trained on the same 3-dataset corpus, so cross-encoder comparisons are corpus-matched.

8.2 Pitfalls (for faithful reproduction)

- **Single-split point estimates are unreliable.** Pointing-raw Spearman has fold-to-fold SD ≈ 0.15 ; earlier single-split numbers were partition artifacts. Use the repeated nested-CV mean, reported as mean $\pm$ SE.
- **Checkpoint identity.** The same encoder appears under two pretraining budgets in our archive; always confirm the checkpoint epoch (and md5) before comparing embeddings.
- **Code/version parity.** Reproduction requires the exact metric MLP, embedding loader, and corrected geodesic function present on the execution host; stale copies change the output.

8.3 Artifacts and policy

Encoder checkpoints (with md5s), the LMA data feeder, the embedding extractor, the nested-CV and baseline harnesses, the perceptual-fine-tune trainer, and the evaluation scripts are archived for end-to-end reproduction, along with the exact per-fold random seeds. The corrected geodesic baseline is computed on raw unpadded sequences (no frame-padding bias). Epoch counts are matched within a backbone family but not forced across families, where an epoch is not a comparable unit.

9 Limitations and scope

- **Pointing's narrower margin and higher variance.** MAMP+pose's raw encoder does beat both geometric baselines on pointing, but by the smallest margin and with the highest fold-to-fold variance; against the supervised TMR baseline the raw encoder is edged on pointing and wins only after the perceptual fine-tune.
- **Three action types.** Generalization to a broader action taxonomy is untested; the per-action heterogeneity we exploit may present differently elsewhere.
- **Rating-set size.** The human rating data ($\sim 1,540$ trials/action) is modest, bounding how much the perceptual fine-tuning can reshape the encoder; a larger set might yield larger or more stable gains.
- **Polarized effort space.** We polarize each bipolar effort factor to $\{-1, 0, +1\}$, so d_{perc} measures dissimilarity among polar-effort combinations, not along the continuous intensity axis; graded-intensity perceptual distance is future work.
- **Learned-baseline conversion.** The TMR comparison requires a BVH $\rightarrow$ SMPL $\rightarrow$ 263-D conversion (≈ 4 cm fit error) that slightly perturbs TMR's input; we frame our result as a conservative lower bound accordingly.